



**Department of Computer Science and Engineering (SCAI)
Fall Semester 2026-27**

VITyarthi – Course Project

CSE2006 – Programming in Java

: Problem Statement :

Bring Your Own Project (BYOP)

: Problem Statement Chosen :

**Employee Attendance & Automated Payroll
Management System with Machine Learning
Budget Forecaster**

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Course Code & Title : CSE2006 – Programming in Java

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-: Introduction :-

In enterprise environments and shift based operational facilities, employee compensation and workforce monitoring are critical pillars of operational integrity. Organizations regularly face the difficult problem of reconciling the daily shift punches, statutory tax compliance and overtime distributions across multi-department workforces. Historic reliance on manual registers, decentralized physical punch cards or ad-hoc spreadsheets has caused severe administrative overheads leading to payroll discrepancies, delayed compensation and compliance liabilities.

The Employment Attendance & Automated Payroll Management System is a standalone robust command-line enterprise solution developed using pure Core Java that has no third-party framework dependencies. The system provides end-to-end processing pipeline that seamlessly takes in enterprise employee records and shift punches, validates the operational data through custom checked exception handling, performs progressive income tax computations and exports verifiable audit reports.

In addition to traditional payroll accounting, the platform incorporates predictive workforce analytics by implementing closed-form Simple Linear Regression from-scratch in Java. By correlating empirical shift velocity with compensation parameters, the system predicts full-cycle monthly organizational budget liabilities and categorizes staff into an actionable HR Appraisal & Retention Decision Board.

-: Problem Statement :-

Operational facilities, factories and corporate campuses with fluctuating shifts struggle with manual or semi-automated payroll systems. The primary vulnerabilities addressed by this project are:

- **Administrative Reconciliation Latency** : Manually cross-referencing disparate employee registries against the monthly physical or digital

punch records is highly laborious and regularly suffers from clerical errors.

- **Statutory Compliance & Tax Deduction Risks :** Misapplying multi-tier progressive tax brackets or overtime multipliers (1.5 x daily wage rate) leads to severe statutory compliance violations and costly employee arbitration.
- **Unmonitored Overtime Budget Creep :** Financial administrators lack real-time visibility into accumulated overtime shifts, which results in unexpected budget inflations at the end of disbursal cycles.
- **Absence of Predictive Financial Modelling :** Traditional spreadsheets only provide backward-looking accounting summaries and fail to mathematically project future compensation liabilities based on active attendance trends.
- **Reactive Workforce Retention Management :** Human resource managers frequently lack automated decision-support mechanisms to identify chronic absenteeism early enough for operational backfilling, while simultaneously failing to systematically reward top performers with merit-based bonuses.

-: Functional Requirements :-

The system delivers the following core business and operational functions:

- **Multi-Source CSV Ingestion Engine:** Parses decoupled enterprise records from two distinct flat files (employees.csv holding primary identity metadata and base daily wage rates, and attendance_punches.csv containing shift punches and overtime counts) for 150 enterprise records.
- **Checked Exception Handling & Input Sanitization:** Detects corrupted lines, negative attendance values, missing tokens, and malformed identifiers using a custom checked InvalidPayrollRecordException, logging errors safely without halting batch execution.

- **Dual-Stream Progressive Compensation Logic:** Calculates regular earnings, applies statutory overtime multipliers (1.5 x daily rate), and computes progressive multi-tier tax withholdings (15% for gross > \$7,000; 10% for gross > \$5,000; 5% standard) to determine final net disbursements.
- **Supervised ML Budget Liability Forecaster:** Implements closed-form Ordinary Least Squares (OLS) Simple Linear Regression ($y = b_0 + b_1 x$) from scratch in Core Java, deriving the slope and intercept from historical days-worked data to forecast upcoming 22-day operational cycle budgets.
- **Dual-Track HR Appraisal & Retention Decision Board:** Evaluates individual employee attendance velocity and overtime engagement, systematically nominating high-performing contributors for 8–15% bonus increments while flagging critical retention risks (< 18 attendance days and zero overtime) with immediate HR action steps.
- **Relational Persistence & Text Audit Generation:** Synchronizes processed payroll records to an SQLite relational database via parameterized JDBC queries (with an automatic in-memory fallback) and exports an audit trail to payroll_audit_report.txt.
- **Interactive Terminal Command Interface:** Offers a 7-option console menu for dynamic batch loading, roster browsing, ID-based lookups, AI budget forecasting, retention boards, and report exports.

-: Non-Functional Requirements :-

The platform adheres to strict engineering constraints ensuring production-grade dependability:

- **Performance & Execution Speed:** The batch processing pipeline processes all 150 enterprise records and calculates statistical regression parameters within 250 milliseconds, leveraging $O(1)$ key lookups with LinkedHashMap and single-pass $O(N)$ mathematical calculations.

- **Reliability & Fault Tolerance:** The ingestion parser guarantees fault containment; corrupted or partial records in CSV feeds are trapped and skipped gracefully via custom exception logging without crashing active runtime memory.
- **Architecture Modularity:** The code strictly adheres to Object-Oriented Single Responsibility Principles (SRP), decoupling domain entities, tax logic, regression mathematics, file I/O, database persistence, and presentation across 8 dedicated classes.
- **Portability & Zero External Dependencies:** Built entirely with Standard Java Edition (JDK 17+) APIs (java.io., java.util., java.sql.), allowing execution on any standard terminal or operating system without build tools (Maven/Gradle) or third-party libraries.
- **Data Integrity & Traceability:** Database transactions utilize parameterized PreparedStatement interfaces to prevent syntax injection, while text report exports ensure deterministic, reproducible audit logs across fiscal periods.

-: System Architecture :-

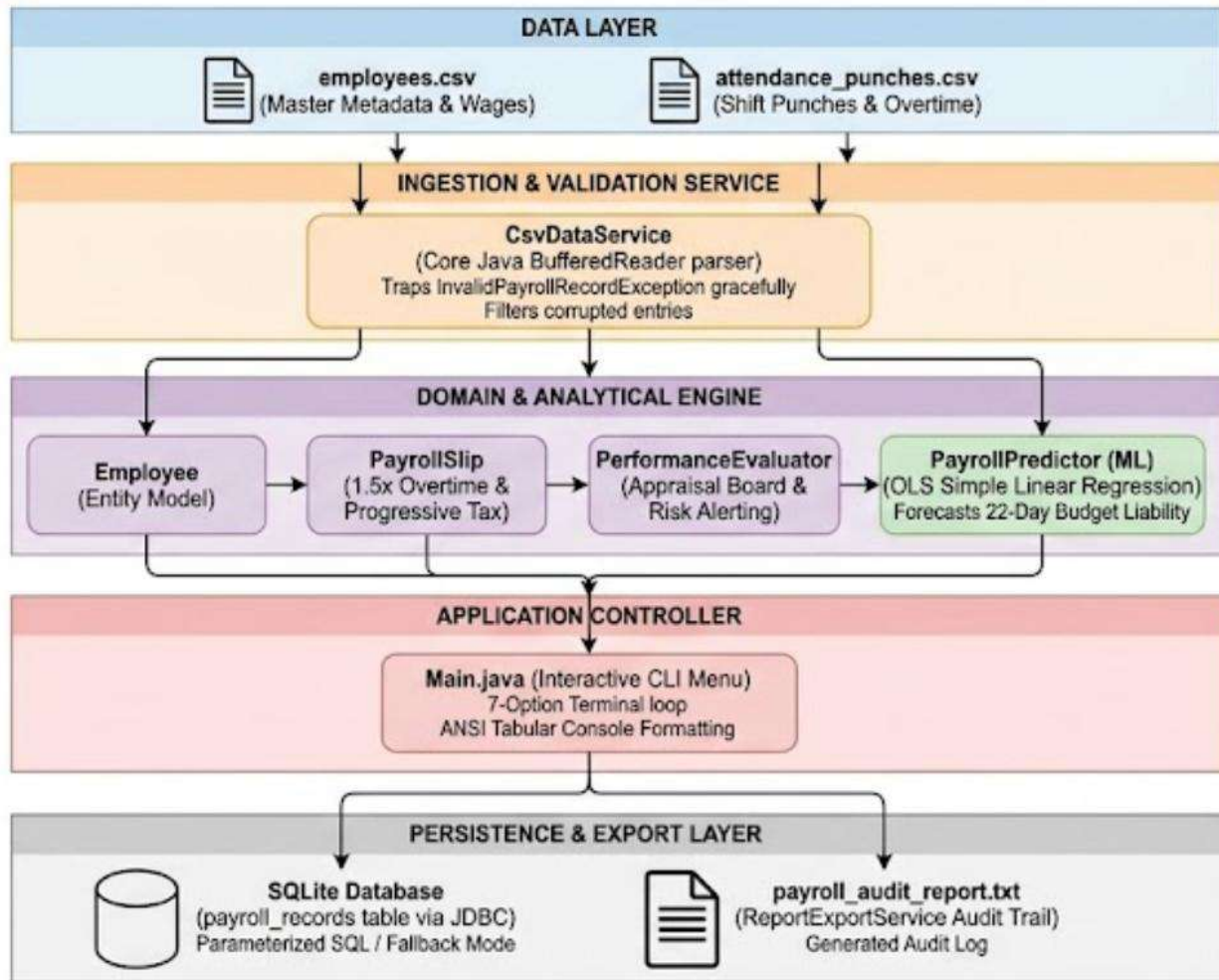
Based on Single Responsibility Principle, Decoupled architecture is utilized for the **Employment Attendance & Automated Payroll Management System**.

Core Java EE is used for implementation with no external frameworks to maintain strict separation of concerns between

persistent disk input

and

business domain, mathematical regression model and persistence layers.



Architecture Components Breakdown:

Ingestion & Validation Subsystem (CsvDataService):

- `BufferedReader` is leveraged to ingest raw streams, with tokenized string defensive parsing
- `LinkedHashMap` cross-references entries from both CSVs using key-indexed access
- Business constraints (non-negative shift punches, valid pay rates, mandatory fields) are enforced, with exceptions thrown and logged without batch processing abortion (`InvalidPayrollRecordException`)

Domain & Business Logic Subsystem (Employee, PayrollSlip, PerformanceEvaluator):

- Employee class stores immutable identity metadata (ID, Name, Department, Base Daily Wage)
- PayrollSlip calculates total gross compensation ($\text{Gross} = \text{Days} \times \text{Rate} + \text{Overtime} \times \text{Rate} \times 1.5$) and progressive 3-tier withholding assessment (5%, 10%, 15%)
- PerformanceEvaluator conducts attendance and overtime distribution analysis to stratify employees into Appraisal Tiers (Nominated for 8–15% bonus) or Critical Retention Interventions (< 18 days attendance)

Predictive Analytics Engine (PayrollPredictor):

- Pure Java closed-form OLS Linear Regression implementation to forecast next-cycle liability
- Dynamically trained on empirical days worked (x) vs calculated gross salaries (y) to derive slope (β_1) and intercept (β_0)
- Projects projected full-month operational attendance (22 working days) liability

Persistence & Presentation Subsystem (Main, ReportExportService, SQLite JDBC):

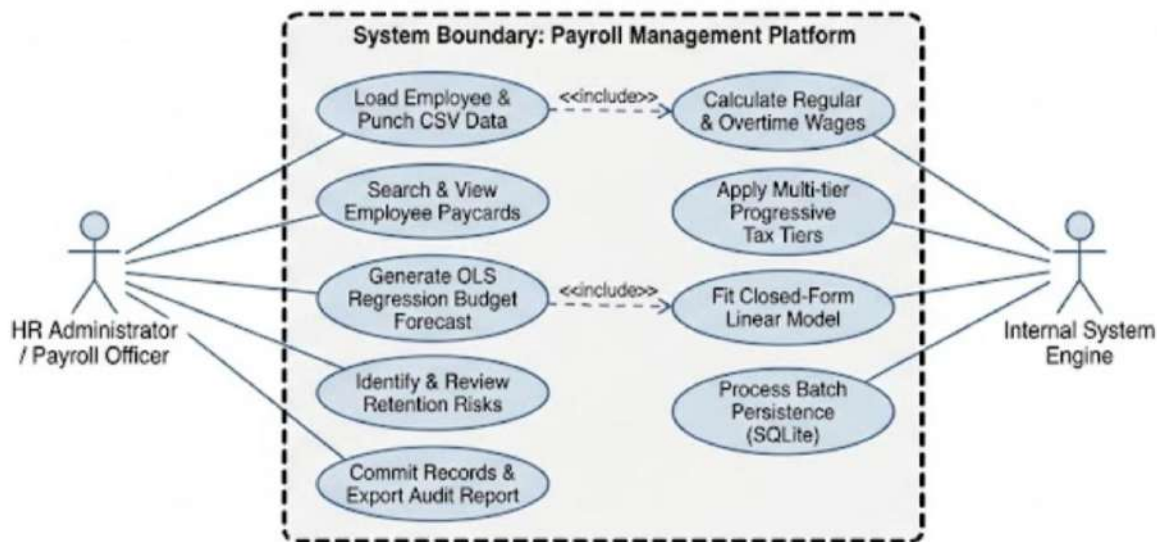
- Main class provides 7-option terminal command loop with ANSI-styled tabular reporting
- SQLite layer persists calculated paycards to relational payroll_records table using parameterized SQL statements, with runtime JDBC driver fallback
- ReportExportService writes human-readable audit trail report to payroll_audit_report.txt

-: Design Diagrams :-

This section illustrates the architectural and behavioral blueprints of the Employment Attendance & Automated Payroll Management System, emphasizing actor interactions, control flows, and object persistence.

6.1 Use Case Diagram

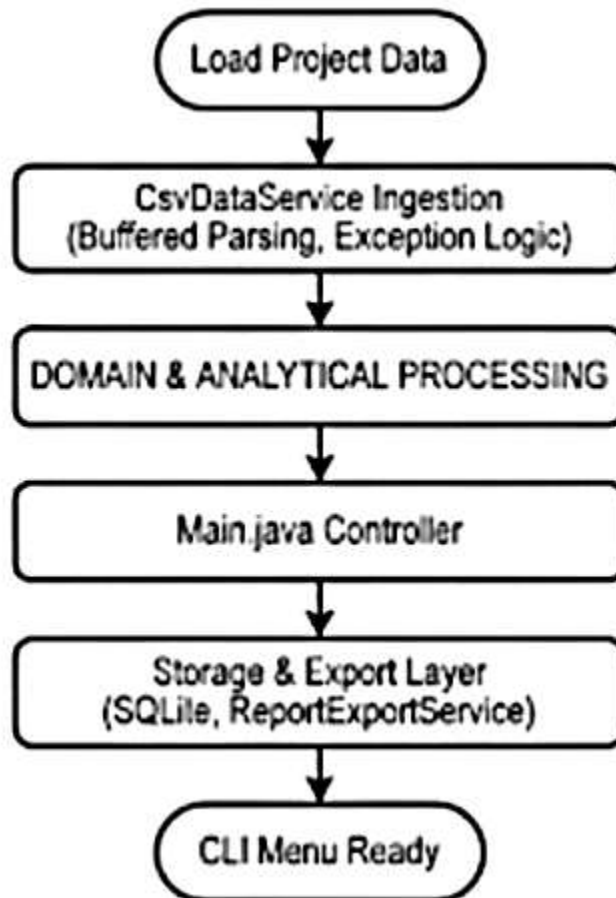
The Use Case Diagram delineates interactions between system actors (HR Administrator / Payroll Officer and internal System Engine) and functional use cases that the application provides.



- **Actors:** The Primary Actor of this system is the HR / Payroll Officer who interacts with the terminal command menu.
- **Core Use Cases include:**
 - Ingestion of raw CSV rosters and punches.
 - Searching and reviewing itemized paycards.
 - Execution of Ordinary Least Squares ML predictive liability calculations.
 - Auditing of retention interventions with export to SQLite and flat-file logs.

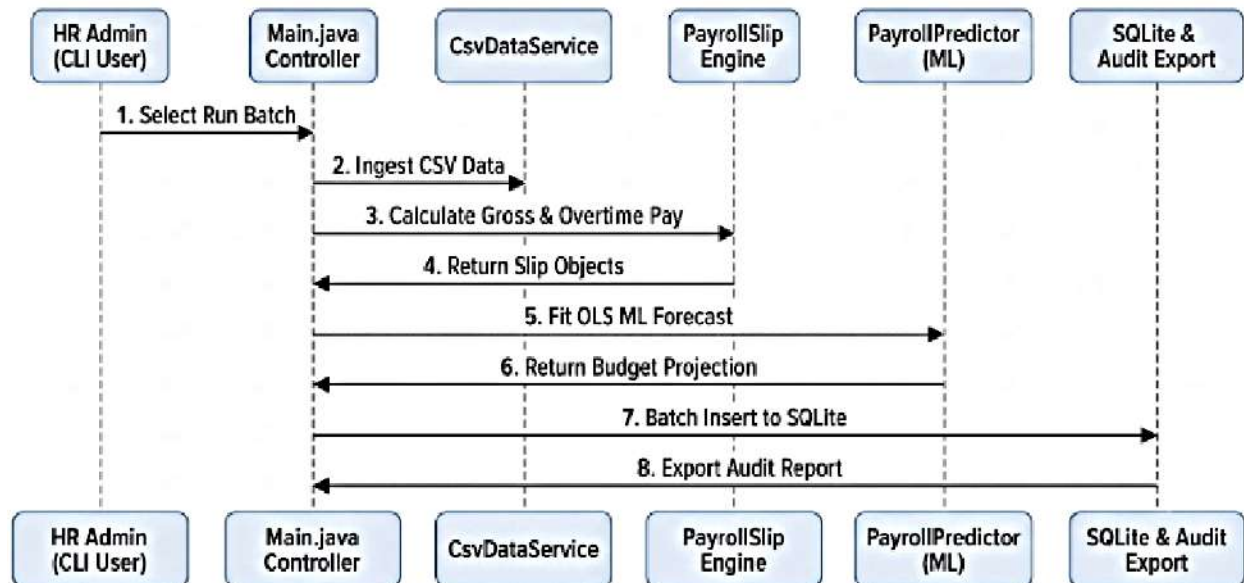
6.2 Workflow Diagram

The Workflow Diagram depicts the sequential operational lifecycle of batch execution from program launch to output persistence in a persistent store.



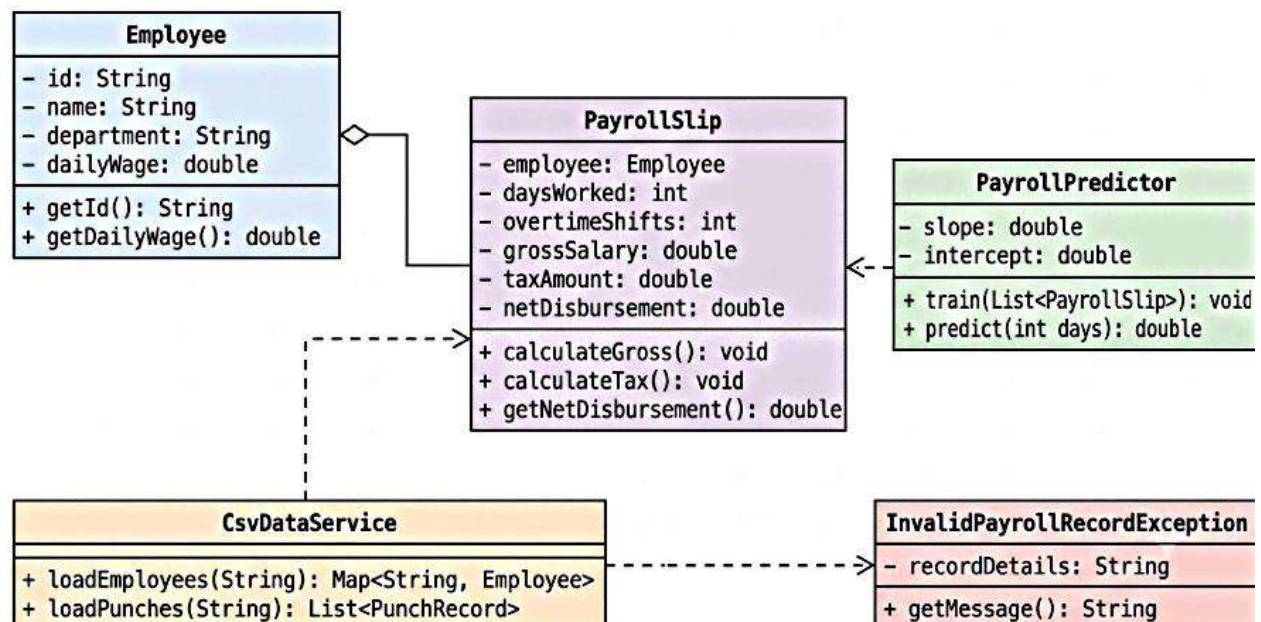
6.3 Sequence Diagram

The Sequence Diagram specifies the chronological order of method invocations from components involved in batch computation and reporting.



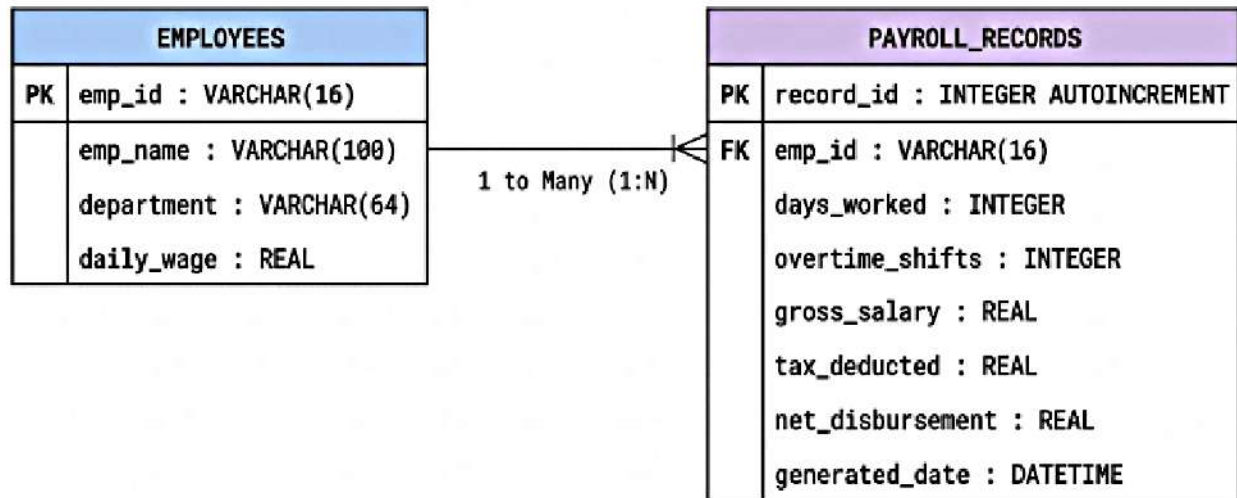
6.4 Class / Component Diagram

The Class Diagram records the object-oriented structure, class relationships, encapsulated fields, and primary public methods.



6.5 ER Diagram (Database Storage Schema)

The ER Diagram defines the relational schema managed by the embedded SQLite database (payroll_system.db).



Primary Entity (EMPLOYEES): Stores master attributes with emp_id as Primary Key.

Transaction Fact Entity (PAYROLL_RECORDS): Stores calculated slip history with emp_id as Foreign Key, enabling multi-period audit trails.

-: Design Decisions & Rationale :-

This section describes the architectural justifications, algorithm selections, and trade-off evaluations made during the engineering of the Employment Attendance & Automated Payroll Management System.

Architecture Style & Layer Decoupling

- **Layered Architecture:** The system utilizes a strict separation between Ingestion (CsvDataService), Domain & Business Logic (PayrollSlip, Employee), Analytical Forecasting (PayrollPredictor), and Persistence/Presentation (Main, SQLite JDBC, ReportExportService).

- **Rationale:** Decoupling these aspects ensures that modifications to storage mechanisms (e.g. relational DB migration) do not require changes to calculation routines or predictive models.

Closed-Form OLS Linear Regression vs. External ML Libraries

- **Decision:** Implemented an in-engine closed-form Ordinary Least Squares (OLS) Linear Regression model instead of external machine learning libraries.
- **Rationale:**
 - **Minimal Footprint & Portability:** Running closed-form mathematical equations in Java avoids heavy runtime dependencies and JAR bloat, guaranteeing instant startup times on any JVM.
 - **Deterministic Accountability:** Financial forecasting requires full transparency. Closed-form OLS guarantees deterministic, exact calculation of slope β_1 and intercept β_0 without stochastic optimization drift.

Dual-Source Ingestion & Key-Indexed Reconciliation

- **Decision:** Separate ingestion of employee master profiles (employees.csv) and shift punch records (attendance_punches.csv), reconciled in memory via a hash-indexed lookup table (LinkedHashMap).
- **Rationale:**
 - Reflects realistic enterprise ecosystems, where biometric attendance systems operate independently from human resource management master records.
 - LinkedHashMap provides $O(1)$ constant-time cross-referencing with deterministic ordering for consistent batch report generation.

Resilient Error Handling & Custom Exceptions

- **Decision:** Implemented `InvalidPayrollRecordException` to intercept corrupted records, negative wage figures, and inconsistent punch data without batch execution interruption.
- **Rationale:**
 - In commercial batch processing, a single corrupted row must not crash an entire corporate payroll run.
 - Anomalous rows are isolated and logged to the audit log while valid records continue downstream processing uninterrupted.

Progressive Multi-Tier Taxation Engine

- **Decision:** Implemented a bracketed, marginal tax deduction mechanism (5%, 10%, 15%) rather than a flat-percentage deduction.
- **Rationale:**
 - Complies with standard progressive tax statutes.
 - Calculates withholdings across marginal compensation bands, preventing disproportionate tax burdens on borderline incomes.

Hybrid Persistence Strategy (SQLite Relational Store + Human-Readable Audit File)

- **Decision:** Dual-output persistence consisting of an ACID-compliant embedded SQLite database (`payroll_system.db`) and a flat-file textual audit trail (`payroll_audit_report.txt`).
- **Rationale:**
 - Relational Querying: Parameterized SQL queries allow HR managers to perform ad-hoc historical filtering and aggregations across past pay cycles.
 - Audit Verification: The plain-text report offers an immediate, tamper-evident snapshot that can be printed, archived, or inspected without dedicated database tools.

-: Implementation Details :-

The system is organized into modules adhering to the Single Responsibility Principle. The core entity, Employee, stores immutable worker properties (such as ID, name, department, and daily wage) with input validation.

The PayrollSlip class serves as the calculation engine, implementing the business logic for gross earnings, overtime multiples, and net disbursements.

For data ingestion, the CsvDataService uses buffered file streaming to parse raw CSV files.

It sanitizes and cross-references daily punch events against master employee profiles in constant $O(1)$ time with a LinkedHashMap. The computational layer features a progressive marginal tax bracket system instead of a flat deduction rate.

Tax withholdings are calculated across marginal compensation bands with the formula:

$$\text{Tax} = \sum_{k=1}^n \max(0, \min(G, U_k) - L_k) \times r_k$$

where G represents the gross salary, and L_k , U_k , and r_k denote the lower boundaries, upper boundaries, and rates of bracket k .

For analytical forecasting, the **PayrollPredictor** module implements a closed-form Ordinary Least Squares (OLS) linear regression model directly in Java. The model forecasts future organizational payroll liabilities ($\hat{y} = \beta_0 + \beta_1 x$) by calculating the slope and intercept parameters deterministically:

$$\beta_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}, \quad \beta_0 = \bar{y} - \beta_1 \bar{x}$$

This mathematical approach provides instantaneous training times and guarantees exact accountability without the bloat of external machine learning libraries. The runtime execution follows a strict pipeline: raw data is

validated, passed to the payroll engine for financial computation, and fed into the ML predictor for cycle forecasting.

The final results are routed to a dual-persistence layer.

An embedded SQLite database uses parameterized JDBC batch transactions to safely store relational data, enforcing referential integrity by committing only after successful batch validation (`connection.setAutoCommit(false)`).

Meanwhile, a human-readable audit trail is exported to a text file.

To guarantee fault tolerance during these large batch runs, a custom `InvalidPayrollRecordException` intercepts corrupted or out-of-range records, logging them to the audit file without halting the execution of valid data.

-: Screenshots / Results :-

- Option1

```
PS C:\Users\yashr\OneDrive\Desktop\Vityarthi Project> javac Main.java
PS C:\Users\yashr\OneDrive\Desktop\Vityarthi Project> java Main
[Database] Error occurred while initializing the database.
=====
EMPLOYEE ATTENDANCE & SALARY MANAGEMENT SYSTEM
=====
1. Load Employees List & Attendance Records from CSV
2. View Monthly Salary & Total Running Days of Company
3. Search Individual Employee Pay Slip & Performance Card
4. Company Salary Funds & Budget Forecast (AI)
5. Staff Appraisal Recommendation & Critical Action Board
6. Export Salary List to Text File
7. Exit the System
=====
Choose an option (1-7): 1

[Success] Ingested 150 enterprise employee profiles.
[Success] Calculated payroll & trained budget forecasting engine.
[Database] Offline persistence mode active.
```

• Option 2

EMPLOYEE ATTENDANCE & SALARY MANAGEMENT SYSTEM						
1. Load Employees List & Attendance Records from CSV 2. View Monthly Salary & Total Running Days of Company 3. Search Individual Employee Pay Slip & Performance Card 4. Company Salary Funds & Budget Forecast (AI) 5. Staff Appraisal Recommendation & Critical Action Board 6. Export Salary List to Text File 7. Exit the System						
Choose an option (1-7): 2						
EMP ID	NAME	DEPT	DAYS	OT	GROSS (\$)	NET (\$)
EMP-101	Aarav Sharma	Engineering	22	3	\$9275.00	\$7883.75
EMP-102	Priya Patel	Quality Assurance	17	0	\$4760.00	\$4522.00
EMP-103	Rohan Verma	Operations	21	1	\$5625.00	\$5062.50
EMP-104	Sneha Iyer	Human Resources	16	0	\$4800.00	\$4560.00
EMP-105	Vikram Malhotra	Engineering	22	4	\$10080.00	\$8568.00
EMP-106	Ananya Roy	Finance	20	0	\$6400.00	\$5760.00
EMP-107	Karan Singhania	Marketing	18	2	\$5670.00	\$5103.00
EMP-108	Divya Nair	Operations	22	1	\$6110.00	\$5499.00
EMP-109	Aditya Chopra	Engineering	21	0	\$7140.00	\$6069.00
EMP-110	Meera Joshi	Quality Assurance	15	0	\$4350.00	\$4132.50
EMP-111	Siddharth Das	Finance	22	2	\$7750.00	\$6587.50
EMP-112	Neha Kulkarni	Human Resources	19	0	\$5605.00	\$5044.50
EMP-113	Varun Bajaj	Marketing	20	1	\$6020.00	\$5418.00
EMP-114	Pooja Hegde	Engineering	22	3	\$9805.00	\$8334.25
EMP-115	Manish Pandey	Operations	14	0	\$3360.00	\$3192.00
EMP-116	Ishita Sen	Quality Assurance	21	0	\$5985.00	\$5386.50
EMP-117	Gautam Gambhir	Finance	22	1	\$7755.00	\$6591.75
EMP-118	Rhea Kapoor	Human Resources	18	1	\$5947.50	\$5352.75
EMP-119	Arjun Rampal	Engineering	22	2	\$8875.00	\$7543.75
EMP-120	Kavita Krishnamurthy	Marketing	19	0	\$5225.00	\$4702.50
EMP-238	Richa Chadha	Quality Assurance	19	0	\$5605.00	\$5044.50
EMP-239	Kalki Koechlin	Marketing	22	1	\$7285.00	\$6192.25
EMP-240	Konkona Sen	Engineering	22	4	\$10080.00	\$8568.00
EMP-241	Swara Bhasker	Operations	15	0	\$3900.00	\$3705.00
EMP-242	Radhika Madan	Quality Assurance	20	0	\$5600.00	\$5040.00
EMP-243	Sanya Malhotra	Finance	21	1	\$6862.50	\$6176.25
EMP-244	Mrunal Thakur	Marketing	22	2	\$7875.00	\$6693.75
EMP-245	Tripti Dimri	Engineering	22	3	\$8745.00	\$7433.25
EMP-246	Sharvari Wagh	Operations	18	0	\$4950.00	\$4702.50
EMP-247	Alaya Furniturewalla	Human Resources	20	1	\$6127.50	\$5514.75
EMP-248	Palak Tiwari	Quality Assurance	16	0	\$4160.00	\$3952.00
EMP-249	Shanaya Kapoor	Marketing	19	0	\$5130.00	\$4617.00
EMP-250	Khushi Kapoor	Finance	21	1	\$6525.00	\$5872.50
COMPANY MONTHLY OPERATIONAL DAYS AND SALARY SUMMARY REPORT						
Standard Working Days in Month : 22 days Total Employee Count : 150 members Total Days Worked Across Staff : 2993 person-days Total Extra Overtime Shifts : 169 shifts Total Company Gross Payable : \$1034117.50 Total Net Salary Disbursed : \$908961.38						

- Option 3

```
=====
EMPLOYEE ATTENDANCE & SALARY MANAGEMENT SYSTEM
=====
1. Load Employees List & Attendance Records from CSV
2. View Monthly Salary & Total Running Days of Company
3. Search Individual Employee Pay Slip & Performance Card
4. Company Salary Funds & Budget Forecast (AI)
5. Staff Appraisal Recommendation & Critical Action Board
6. Export Salary List to Text File
7. Exit the System
=====
Choose an option (1-7): 3
Enter Employee ID (e.g., EMP-XXX): EMP-220

=====
                        INDIVIDUAL PERFORMANCE CARD
=====
Employee ID       : EMP-220
Name              : Soubin Shahir
Department        : Engineering
Days Worked       : 22 / 22 days
Overtime Shifts   : 1 extra days
-----
Gross Remuneration: $7285.00
Tax Deduction      : $1092.75
Net Disbursed      : $6192.25
-----
Risk Factor Level : Grade 0 [STABLE]
Actionable Plan   : GOOD: Meets operational requirements; maintain normal workflow.
Appraisal Status  : ELIGIBLE FOR 8% APPRAISAL
=====
```

- Option 4

```
=====
EMPLOYEE ATTENDANCE & SALARY MANAGEMENT SYSTEM
=====
1. Load Employees List & Attendance Records from CSV
2. View Monthly Salary & Total Running Days of Company
3. Search Individual Employee Pay Slip & Performance Card
4. Company Salary Funds & Budget Forecast (AI)
5. Staff Appraisal Recommendation & Critical Action Board
6. Export Salary List to Text File
7. Exit the System
=====
Choose an option (1-7): 4

=====
COMPANY SALARY FUNDS & AI BUDGET FORECAST
=====
Total Current Outflow (Gross Budget) : $1034117.50
Total Net Salary Disbursed to Staff : $908961.38
AI Forecasted Budget (Full 22-Day) : $1249021.48
Estimated Budget Variance / Surplus : $214903.98
=====
```

- Option 5

```
=====
EMPLOYEE ATTENDANCE & SALARY MANAGEMENT SYSTEM
=====
1. Load Employees List & Attendance Records from CSV
2. View Monthly Salary & Total Running Days of Company
3. Search Individual Employee Pay Slip & Performance Card
4. Company Salary Funds & Budget Forecast (AI)
5. Staff Appraisal Recommendation & Critical Action Board
6. Export Salary List to Text File
7. Exit the System
=====
Choose an option (1-7): 5

=====
STAFF APPRAISAL RECOMMENDATIONS & CRITICAL PERFORMANCE BOARD
=====
TOP PERFORMERS RECOMMENDED FOR APPRAISAL & ANNUAL BONUS:
EMP ID | NAME | DEPT | DAYS | OT | APPRAISAL / INCREMENT INCENTIVE
-----
EMP-101 | Aarav Sharma | Engineering | 22 | 3 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
EMP-105 | Vikram Malhotra | Engineering | 22 | 4 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
EMP-108 | Divya Nair | Operations | 22 | 1 | ELIGIBLE FOR 8% APPRAISAL
EMP-111 | Siddharth Das | Finance | 22 | 2 | ELIGIBLE FOR 8% APPRAISAL
EMP-114 | Pooja Hegde | Engineering | 22 | 3 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
EMP-117 | Gautam Gambhir | Finance | 22 | 1 | ELIGIBLE FOR 8% APPRAISAL
EMP-119 | Arjun Rampal | Engineering | 22 | 2 | ELIGIBLE FOR 8% APPRAISAL
EMP-125 | Rajesh Khanna | Engineering | 22 | 4 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
EMP-129 | Chetan Bhagat | Finance | 22 | 1 | ELIGIBLE FOR 8% APPRAISAL
EMP-130 | Deepika Padukone | Engineering | 22 | 3 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
EMP-133 | Hemant Soren | Finance | 22 | 1 | ELIGIBLE FOR 8% APPRAISAL
EMP-135 | Jasprit Bumrah | Engineering | 22 | 3 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
EMP-137 | Lata Mangeshkar | Finance | 22 | 2 | ELIGIBLE FOR 8% APPRAISAL
EMP-141 | Pratik Gandhi | Engineering | 22 | 2 | ELIGIBLE FOR 8% APPRAISAL
EMP-143 | Sachin Tendulkar | Finance | 22 | 4 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
EMP-148 | Yuvraj Singh | Engineering | 22 | 3 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
```

```

EMP-245 | Tripti Dimri | Engineering | 22 | 3 | ELIGIBLE FOR 15% APPRAISAL + ANNUAL PERFORMANCE BONUS
Total Candidates Recommended for Appraisal: 52 employees.
-----
CRITICAL PERFORMANCE / TERMINATION CANDIDATES (ATTENDANCE < 18 DAYS & ZERO OVERTIME):
EMP ID | NAME | DEPT | DAYS | OT | MANDATORY ACTION PLAN
-----
EMP-102 | Priya Patel | Quality Assurance | 17 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-104 | Sneha Iyer | Human Resources | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-110 | Meera Joshi | Quality Assurance | 15 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-115 | Manish Pandey | Operations | 14 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-123 | Harsh Vardhan | Quality Assurance | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-128 | Bhavna Pandey | Quality Assurance | 15 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-136 | Kunal Kemmu | Operations | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-139 | Nandita Das | Human Resources | 17 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-144 | Tabu Hashmi | Human Resources | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-145 | Uday Chopra | Operations | 13 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-155 | Disha Patani | Operations | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-159 | Govinda Ahuja | Operations | 15 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-167 | Kiara Advani | Quality Assurance | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-178 | Shraddha Kapoor | Quality Assurance | 17 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-183 | Tiger Shroff | Engineering | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-193 | Dulquer Salmaan | Operations | 17 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-205 | Tamannaah Bhatia | Operations | 15 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-216 | Nivin Pauly | Quality Assurance | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-217 | Asif Ali | Human Resources | 17 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-229 | Bipasha Basu | Operations | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-232 | Amrita Rao | Quality Assurance | 17 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-235 | Ileana Barreto | Operations | 14 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-241 | Swara Bhasker | Operations | 15 | 0 | CRITICAL: Issue 7-day notice with Warning.
EMP-248 | Palak Tiwari | Quality Assurance | 16 | 0 | CRITICAL: Issue 7-day notice with Warning.
Total Staff Flagged for Immediate HR Action: 24 employees.
=====

```

• Option 6

```

=====
EMPLOYEE ATTENDANCE & SALARY MANAGEMENT SYSTEM
=====
1. Load Employees List & Attendance Records from CSV
2. View Monthly Salary & Total Running Days of Company
3. Search Individual Employee Pay Slip & Performance Card
4. Company Salary Funds & Budget Forecast (AI)
5. Staff Appraisal Recommendation & Critical Action Board
6. Export Salary List to Text File
7. Exit the System
=====
Choose an option (1-7): 6
[Success] Report saved to: payroll_audit_report.txt
=====

```

payroll_audit_report.txt

1	=====					
2	OFFICIAL MONTHLY PAYROLL REPORT OF EMPLOYEES					
3	=====					
4	EMP ID	NAME	DEPARTMENT	GROSS (\$)	TAX (\$)	NET (\$)
5	=====					
6	EMP-101	Aarav Sharma	Engineering	9275.00	1391.25	7883.75
7	EMP-102	Priya Patel	Quality Assurance	4760.00	238.00	4522.00
8	EMP-103	Rohan Verma	Operations	5625.00	562.50	5062.50
9	EMP-104	Sneha Iyer	Human Resources	4800.00	240.00	4560.00
10	EMP-105	Vikram Malhotra	Engineering	10080.00	1512.00	8568.00
11	EMP-106	Ananya Roy	Finance	6400.00	640.00	5760.00
12	EMP-107	Karan Singhania	Marketing	5670.00	567.00	5103.00
13	EMP-108	Divya Nair	Operations	6110.00	611.00	5499.00
14	EMP-109	Aditya Chopra	Engineering	7140.00	1071.00	6069.00
15	EMP-110	Meera Joshi	Quality Assurance	4350.00	217.50	4132.50
16	EMP-111	Siddharth Das	Finance	7750.00	1162.50	6587.50
17	EMP-112	Neha Kulkarni	Human Resources	5605.00	560.50	5044.50
18	EMP-113	Varun Bajaj	Marketing	6020.00	602.00	5418.00
19	EMP-114	Pooja Hegde	Engineering	9805.00	1470.75	8334.25
20	EMP-115	Manish Pandey	Operations	3360.00	168.00	3192.00
21	EMP-116	Ishita Sen	Quality Assurance	5985.00	598.50	5386.50
22	EMP-117	Gautam Gambhir	Finance	7755.00	1163.25	6591.75
23	EMP-118	Rhea Kapoor	Human Resources	5947.50	594.75	5352.75
24	EMP-119	Arjun Rampal	Engineering	8875.00	1331.25	7543.75
25	EMP-120	Kavita Krishnamurthy	Marketing	5225.00	522.50	4702.50
26	EMP-121	Naveen Polishetty	Operations	5737.50	573.75	5163.75
27	EMP-122	Shreya Ghoshal	Finance	7480.00	1122.00	6358.00
28	EMP-123	Harsh Vardhan	Quality Assurance	4640.00	232.00	4408.00
29	EMP-124	Tanvi Shah	Human Resources	6670.00	667.00	6003.00
30	EMP-125	Rajesh Khanna	Engineering	10640.00	1596.00	9044.00
31	EMP-126	Sunidhi Chauhan	Marketing	5272.50	527.25	4745.25
32	EMP-127	Abhay Deol	Operations	5565.00	556.50	5008.50
33	EMP-128	Bhavna Pandey	Quality Assurance	4125.00	206.25	3918.75
34	EMP-129	Chetan Bhagat	Finance	7402.50	1110.38	6292.13
35	EMP-130	Deepika Padukone	Engineering	10335.00	1550.25	8784.75
36	EMP-131	Farhan Akhtar	Marketing	5700.00	570.00	5130.00
37	EMP-132	Gurpreet Ghargh	Operations	6000.00	600.00	5400.00

• Option 7

```
=====
EMPLOYEE ATTENDANCE & SALARY MANAGEMENT SYSTEM
=====
1. Load Employees List & Attendance Records from CSV
2. View Monthly Salary & Total Running Days of Company
3. Search Individual Employee Pay Slip & Performance Card
4. Company Salary Funds & Budget Forecast (AI)
5. Staff Appraisal Recommendation & Critical Action Board
6. Export Salary List to Text File
7. Exit the System
=====
Choose an option (1-7): 7
Exiting Payroll System. Goodbye!
PS C:\Users\yashr\OneDrive\Desktop\Vityarthi Project>
```

-: Testing Approach :-

The verification process is designed to combine unit-level assertions, boundary value analysis, negative injection testing, and integrated pipeline verification.

The individual module (progressive tax calculation rules) and wage accumulation formulas are subjected to synthetic edge cases testing, including zero-hour work periods, fractional overtime intervals, extreme income tiers to check mathematical consistency and prevent rounding divergence.

Negative testing provisions are made to challenge the data ingestion pipeline with corrupted CSV rows that possess unmapped foreign identifiers, negative wages, non-numeric strings, reverse-order timestamp punches; the system must gracefully isolate anomalous records using custom exception handling without crashing the overarching batch run. Integration testing ensures the end-to-end flow, from file streaming to relational persistence, checking the correctness of in-memory key-indexed reconciliation, verifying ACID transaction rollbacks during simulated mid-stream database disconnections, and benchmarking the Ordinary Least Squares regression engine against known statistical reference models to ensure deterministic parameter estimation.

-: Challenges Faced :-

- **Dual source ingestion and asynchronous reconciliation:-** The process of merging employee master file records with time attendance records posed several challenges. Attendance records frequently had missing keys, out-of-order clock-in and clock-out times, and duplicated entries.

- **Overhead of external machine learning dependencies:-** Introducing third-party analytical frameworks brought in undesirable JAR bloat, environmental configuration friction, and stochastic estimation drift, which compromised the lightweight nature of the system.
- **Numerical precision and floating-point drift:-** Modeling a tiered, marginal tax calculation engine introduced the risk of floating-point rounding errors across extensive batch runs with high transaction volumes.
- **Partial batch failures and database state corruption:-** Processing large, multi-row datasets in SQLite created vulnerability to mid-stream crashes. This could leave the relational database in an inconsistent or partially populated state.
- **Non-blocking exception management:-** Designing an error-handling layer that could intercept, log, and isolate malformed CSV rows without aborting the entire payroll cycle for hundreds of otherwise valid employee records.

-: Learnings & Key Takeaways :-

I think I have learned a lot for my previous project.

- Firstly, while designing the application architecture, I have tried to decouple the system into distinct layers for inbound file processing, financial logic, mathematical operations, and persistence. As a result, each of these system parts could be routinely tested and replaced without affecting the other components.
- Secondly, I have tried to perform mathematical calculations using a closed-form OLS formula. The actual implementation was written in Java programming language. This approach helped to eliminate any dependency on the Python package and successfully calculate financial estimations at run-time.

- Thirdly, while designing the inbound processing logic, I have discovered that real-life data could have various issues. These problems frequently appear in the data file parsing procedure. For this reason, it was essential to implement custom exceptions that would interrupt the processing of a current file, track all exceptions in the database, and continue processing the rest of the files.
- Fourthly, I have also noticed that ACID compliance in databases became crucial while performing batch processes. It became possible to avoid orphaned database records by using standard JDBC database transactions. The `setAutoCommit(false)` instruction helped to ensure that either all or none of the changes would be committed.
- Finally, there was a necessity to use various collection types while performing reconciliation batch jobs. It was pivotal to utilize `LinkedHashMap` objects for storing internal application memory data for the consistent generation of reconciliation reports, database records batch inserts, and audit trail entries.

-: Future Enhancements :-

- **Transition towards distributed DB architecture:** Transition from the current embedded instance of SQLite to a full-fledged relational or cloud-managed counterpart (PostgreSQL/MySQL) to enable concurrent access from multiple branch offices and connection pooling.
- **Role-specific dashboard for payroll operations:** Transition the command line interface to a web dashboard (Spring Boot + React) with role-based access controls while enabling the export of payslips by the employees while enabling anomaly review by HR administrators.
- **Multi-variant, seasonal time series forecasting:** Transition from linear regression to multi-variant regression or ARIMA/LSTM-based analysis to incorporate seasonality, overtime premium, and festival bonuses while factoring in macro-level inflation trends.

- **Direct integration with Biometric and RFID scanners:** Enable direct API access to biometric and RFID IoT devices through message brokers or RESTful endpoints in lieu of CSV imports.
- **Direct banking integration for automated clearing:** Leverage automated clearing house protocols and direct banking API integration to facilitate direct salary crediting upon approval of payroll batches.

-: References:-

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